Hilbert's Sixth Problem*

Ian Coley

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This talk will come in four parts. First, I will introduce the Riemann hypothesis (RH) as it was first introduced, using the Riemann zeta function, and discuss briefly its connection to number theory and prime numbers. Second, I will make this connection explicit by discussion a famous problem equivalent to RH. Third, I will examine the generalised Riemann hypothesis (GRH) and, finally, will list some other interesting things implied by RH and GRH.

Parts of this talk were taking from Wikipedia, Terry Tao's Wordpress [1], a paper by Thomas Wright [3], MathOverflow [2], and the official problem statement of the Riemann Hypothesis from the Clay Mathematics Institute written by E. Bombieri [4].

1 What is RH?

To begin, we can talk about the function

$$Z(x) = \sum_{n=1}^{\infty} \frac{1}{n^x}.$$

For now, we can assume this is a function on $\mathbb{R}$ to $\mathbb{R}$. This function has some interesting values for some small positive x:

$$Z(2) = \frac{\pi^2}{6}, \quad Z(4) = \frac{\pi^4}{90}.$$

Euler proved both of these identities. There is an obvious problem with Z as written: if we try to do $Z(x)$ for $x \leq 1$, we get a function that goes to infinity. However, we can extend Z to a complex function defined on the half-plane of $\mathbb{C}$ given by $\Re(x) > 1$: if $s = a + bi$,

$$\begin{aligned} \left| \sum_{n=1}^{\infty} \frac{1}{n^x} \right| &\leq \sum_{n=1}^{\infty} \left| \frac{1}{n^a \cdot n^{bi}} \right| \\ &= \sum_{n=1}^{\infty} \frac{1}{n^a}, \text{ which converges.} \end{aligned}$$

*The title of this talk was supposed to be Hilbert's Eighth Problem. Hilbert's Sixth Problem is what went up on the UCLA Math Department website, so I honour the error here as well.

We get the connection to number theory even without traveling to complex analysis yet. Another way to write $Z(x)$ is via an infinite product instead of an infinite sum, due to Euler in 1748:

$$\begin{aligned} Z(x) &= \left(\frac{1}{1-\frac{1}{2^x}}\right)\left(\frac{1}{1-\frac{1}{3^x}}\right)\left(\frac{1}{1-\frac{1}{5^x}}\right)\cdots \\ &= \prod_{p \text{ prime}} \frac{1}{1-\frac{1}{p^x}}. \end{aligned}$$

This equivalent statement comes from the following idea: using that

$$\frac{1}{1-\frac{1}{p^x}} = 1 + \frac{1}{p^x} + \frac{1}{(p^x)^2} + \cdots,$$

we see that the terms of this (so-called) Euler product will be of the form

$$\frac{1}{(p_1^x)^{a_1}} \cdots \frac{1}{(p_m^x)^{a_m}} = \frac{1}{(p_1^{a_1} \cdots p_m^{a_m})^x}$$

for each possible combination of the p_i and $a_i \in \mathbb{N}$. But each possible combination of the a_i and p_i gives you a unique natural number n because of unique factorisation of the integers. This (verbally) shows that we are talking about the same power series.

The reason that Riemann gets his name all over this is the rephrasing of $Z(x)$ in terms of a complex function on all of $\mathbb{C}$. He defined the complex function

$$\zeta(s) = \frac{1}{1-2^{1-s}} \sum_{n=0}^{\infty} \frac{1}{2^{n+1}} \sum_{k=0}^{n} (-1)^k \binom{n}{k} (k+1)^{-s}.$$

This function agrees with $Z(x)$ for $\Re(x) > 1$ and, further, it can be analytically continued to a meromorphic function on all of $\mathbb{C}$ with a simple pole at $s = 1$ of residue 1 via the (symmetric) functional equation

$$\pi^{-s/2}\,\Gamma\left(\frac{s}{2}\right)\zeta(s) = \pi^{-(1-s)/2}\,\Gamma\left(\frac{1-s}{2}\right)\zeta(1-s).$$

For good measure, there is another functional equation:

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right)\Gamma(1-s)\zeta(1-s). \tag{1}$$

Riemann then defined a new function ξ defined by

$$\xi(t) = \frac{1}{2}\, s(s-1)\pi^{-s/2}\,\Gamma\left(\frac{s}{2}\right)\zeta(s),$$

where $s = 1/2 + it$. Riemann proved that $\xi(t)$ is an even entire function of t whose zeroes have imaginary part between $-i/2$ and $i/2$. A zero α of $\xi(t)$ corresponds to a *nontrivial* zero $1/2 + i\alpha$ of $\zeta(s)$. The trivial zeroes of $\zeta(s)$ are the negative even integers $-2, -4, \ldots$. You can see this from Equation 1 given above. The Riemann hypothesis is that $\xi(t)$ has only real zeroes, hence:

Conjecture (Riemann, 1859). The nontrivial zeroes of $\zeta(s)$ have real part equal to $\frac{1}{2}$.

This is/was Hilbert's 8th problem, later becoming a Millennium Prize Problem. Lest it remain unsaid, this problem is not solved.

2 Famous implications of RH

The most famous (arguably) implication of RH is a sharp bound on the error implied by the Prime Number Theorem. Recall the following:

Theorem (PNT). Let $\pi(x)$ be the number of prime numbers less than or equal to x. Then

$$\lim_{x\to\infty} \frac{\pi(x)}{x/\log x} = 1.$$

We can put this another way: $\pi(x)$ is approximately $\mathrm{Li}(x)$, where

$$\mathrm{Li}(x) = \int_2^x \frac{dt}{\log t}.$$

In 1899, La Vallée Poussin proved that

$$\pi(x) = \mathrm{Li}(x) + O\left(x^{-a\sqrt{\log x}}\right),$$

for some positive constant a, which gives a much better estimate on the error between $\mathrm{Li}(x)$ and $\pi(x)$. However, we can do much better:

Theorem (Helge von Koch, 1901). $\pi(x) = \mathrm{Li}(x) + O(\sqrt{x}\log x)$ if and only if the Riemann hypothesis is true.

The is the same Koch as the eponymous snowflake. The fact that this is an equivalence is quite mysterious. Right now we will explore the converse direction.

We define $\Lambda(n)$, the (Hans) von Mangoldt function, by

$$\Lambda(n) = \begin{cases} \log p & n = p^k \text{ for some prime } p,\ k \geq 1 \\ 0 & \text{otherwise.} \end{cases}$$

This function satisfies the identity

$$\log n = \sum_{d|n} \Lambda(d).$$

We treat both sides like the coefficients of a Dirichlet series, i.e. a series of the form

$$\sum_{n=1}^{\infty} \frac{a_n}{n^s},$$

where $a_n \in \mathbb{C}$ is a complex series, then we obtain the equality

$$\sum_{n=1}^{\infty} \frac{\log n}{n^s} = \sum_{n=1}^{\infty} \sum_{d|n} \frac{\Lambda(d)}{n^s}.$$

If we write $n = dm$, then the right hand side becomes

$$\sum_{n=1}^{\infty} \sum_{d|n} \frac{\Lambda(d)}{n^s} = \left(\sum_{d=1}^{\infty} \frac{\Lambda(d)}{d^s} \right) \left(\sum_{m=1}^{\infty} \frac{1}{m^s} \right) = \zeta(s) \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s}.$$

The lefthand side, meanwhile, is formally the negative derivative of the Riemann zeta function $-\zeta'(s)$. This gives us the identity

$$\sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s} = -\frac{-\zeta'(s)}{\zeta(s)}.$$

This is highly reminiscent of logarithmic differentiation, so integrating both sides we get

$$\sum_{n=1}^{\infty} \frac{\Lambda(n)}{\log n} n^{-s} = \log \zeta(s) \implies \zeta(s) = \exp \left(\sum_{n=1}^{\infty} \frac{\Lambda(n)}{\log n} n^{-s} \right).$$

Now, we want to express $\zeta(s)$ using the Weierstrass factorisation theorem, which gives a formulation of a a meromorphic function in terms of a product of its poles and zeroes (as in the fundamental theorem of algebra). That is,

$$\zeta(s) = \frac{1}{s-1} \cdot \prod_{\rho : \zeta(\rho) = 0} (s - \rho) \cdots$$

where there is some more to multiply on the right side that is going to be handwaved, because it doesn't matter for the asymptotics. If we take the logarithm of this, we get

$$\sum_{n=1}^{\infty} \frac{\Lambda(n)}{\log n} n^{-s} = -\log(s-1) + \sum_{\rho} \log(s - \rho) + \cdots$$

Notice that

$$\frac{1}{s-\rho} = \int_1^{\infty} t^{\rho - s} \frac{dt}{t} \implies \log(s-\rho) = -\int_1^{\infty} t^{\rho - s - 1} \frac{dt}{\log t}$$

when we integrate with the respect to s. This gives us the heuristic

$$\log(s-\rho) \sim -\sum_{n=1}^{\infty} \frac{n^{\rho - s - 1}}{\log n}.$$

If we compare this with what we had above, we get

$$\sum_{n=1}^{\infty} \frac{\Lambda(n)}{\log n} n^{-s} \sim -\log(s-1) + \sum_{\rho} \sum_{n=1}^{\infty} \frac{n^{\rho - s - 1}}{\log n}.$$

This yields

$$\Lambda(n) \sim 1 - \sum_{\rho} n^{\rho - 1}.$$

This can be interpreted through some complicated means into

$$\sum_{n \leq x} \Lambda(n) \sim x - \sum_{\rho} \frac{x^{\rho}}{\rho}.$$

This becomes more rigorous after some more work and we are able to conclude that, if RH holds, we have

$$\sum_{n \leq x} \Lambda(n) = x + O(\sqrt{x}(\log x)^2).$$

Using summation by parts, one can show that this is the same as

$$\pi(x) = \sum_{p \leq x} 1 = \mathrm{Li}(x) + O(\sqrt{x} \log x)$$

since $\mathrm{Li}(x)$ and $x/\log x$ are approximately equal. The fact that this bound implies the Riemann hypothesis is more complicated and will not be done here.

3 The Generalised Riemann Hypothesis

$\zeta(s)$ is only one of a larger family of functions called L-functions. These are often used to get arithmetic data from algebraic objects. A useful family is that of *Dirichlet L-functions*, introduced first by Dirichlet in 1837 in his proof that any arithmetic progression contains infinitely many prime numbers. These L-functions are defined by:

$$L(s, \chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s}$$

for $\Re(s) > 1$. χ here is a *Dirichlet character*, which arise from completely multiplicative characters on $\mathbb{Z}/k\mathbb{Z}$ for some k. There is an Euler product form of $L(s, \chi)$:

$$L(s, \chi) = \prod_{p \text{ prime}} (1 - \chi(p)p^{-s})^{-1}.$$

This can be analytically continued into a meromorphic function on the whole complex plane. The generalised Riemann hypothesis is that they do satisfy a property similar to that which $\zeta(s)$ should satisfy, namely:

Conjecture (GRH). If $L(\chi, s) = 0$, and $\Re(s) \in (0, 1)$, then $\Re(s) = 1/2$.

The Dedekind zeta function for a number field K is defined by

$$\zeta_K(s) = \sum_{\mathfrak{a} \subset \mathcal{O}_K} \frac{1}{(N_{K/\mathbb{Q}}(\mathfrak{a}))^s},$$

where $N_{K/Q}$ is the absolute norm of $\mathfrak{a}$ (i.e. the index of $\mathfrak{a}$ in $\mathcal{O}_K$), $\mathfrak{a}$ is a nonzero ideal, and $\Re(s) > 1$. If $K = \mathbb{Q}$, this reduces to the Riemann zeta function. This also has a Euler product expression, as rings of algebraic integers are Dedekind domains:

$$\zeta_K(s) = \prod_{\mathfrak{p} \subset \mathcal{O}_K} (1 - (N_{K/\mathbb{Q}}(\mathfrak{p}))^{-s})^{-1},$$

where $\mathfrak{p}$ is a prime ideal of $\mathcal{O}_K$. In the same way as above, we can analytically continue $\zeta_K(s)$ to a meromorphic function on all of $\mathbb{C}$ which should have nontrivial zeroes only at points with $\Re(s) = 1/2$. Wikipedia calls this the *extended* RH, but we can keep it in the same family as GRH.

Unfortunately, we have thus far not found a single example of validity or failure of a Riemann hypothesis for anything, and the problem does not seem more or less tractable in this case.

4 Some other implications

For as much time as remains, I will now list some other things implied by RH or some GRH.

All numerical data shows that $\pi(x) < \mathrm{Li}(x)$. In 1914, Littlewood used the usual RH to show that the inequality reverses infinitely often. It was proved unconditionally in 1955 by Skewes that the inequality reverses for some $x < 10^{10^{10^{963}}}$, though in 1933 he used RH to show that this occurs for some $x < 10^{10^{10^{34}}}$.

In 1919, Cramér showed that RH implies that the prime gap $p_{k+1} - p_k = O(\sqrt{p_k} \log p_k)$. However, Cramér conjectured that the prime gap is really $O((\log p_k)^2)$, which is a much better gap than the one given. Numerical data supports this.

In 1853, Chebyshev proved that, for large x, the number of primes less than x equal to 3 mod 4 is usually greater than the number of those equal to 1 mod 4. He conjectured that this was always true, and that in fact

$$\lim_{x \to 1^-} \sum_{p \neq 2} (-1)^{(p+1)/2} x^p = \infty,$$

i.e., the number of primes 3 mod 4 is 'greater'. Littlewood in 1914 showed that in the 'prime race', the winner changes infinitely often. Regardless, in 1917, Hardy-Littlewood and Landau (independently) showed that this Chebyshev's conjecture is equivalent to a GRH for the L-function of the nontrivial character obtained from $\mathbb{Z}/4\mathbb{Z}$.

The 'odd' Goldbach conjecture (1742) says that all odd integers $n \geq 7$ are the sum of three primes. In 1923, Hardy and Littlewood shows that if all Dirichlet L-functions have no zeroes in $\mathbb{R}(s) > 3/4 + \varepsilon$ for some fixed ε, then the odd Goldbach conjecture is true for sufficiently large n. In 1937, Vinogradov proved this fact unconditionally, so removed the need for GRH. However, in 1997, Deshouillers, Effinger, the Riele, and Zinoviev proved that GRH implies that the odd Goldbach conjecture is true.

In 1973, Weinberger shows that if GRH is true for Dedekind zeta functions, then any number field with an infinite unit group (i.e. $K \neq \mathbb{Q}, \mathbb{Q}[\sqrt{-D}]$ is Euclidean if it has class number 1. As a specific case, if $D > 0$ is a squarefree positive integer, then $\mathbb{Z}[\sqrt{D}]$ is a unique factorisation domain if and only if it is Euclidean.

Gauss in 1801 conjectured that there are only finitely many quadratic fields with class number 1. In 1913, Gronwall showed this is true if the L-functions of all imaginary quadratic Dirichlet characters have no zeroes in some common strip $1 - \varepsilon < \Re(s) < 1$. In 1933, Deuring and Mordell showed Gauss' conjecture is true if the ordinary RH is *false*, and in 1934 Heilbronn showed Gauss' conjecture if true if GRH is *false* for some Dirichlet L-function

of an imaginary quadratic character. Therefore Gauss' conjecture is true. In 1935, Siegel proved Gauss' conjecture is true without referring to GRH.

Euler called an integer $n \geq 1$ *convenient* if any odd integer greater than 1 that has a unique representation as $x^2 + ny^2$ in positive integers x, y with $\gcd(x, ny) = 1$ is a prime number. Euler found 65 convenient numbers below 10000. In 1934, Chowla showed that there are finitely many convenient numbers. In 1973, Weinberger showed there is at most one convenient number not in Euler's list, and if the L-functions of all quadratic Dirichlet characters satisfy GRH then Euler's list is complete.

References

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[4] *Problems of the Millennium: the Riemann Hypothesis.*
`http://www.claymath.org/sites/default/files/official_problem_description.pdf`